

LIFE CYCLE ASSESSMENT REPORT

RCC Residential Tower with Shear Wall Core
Chandigarh, India | B+G+15 Floors

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Executive Summary

This Life Cycle Assessment (LCA) evaluates the environmental impact of a B+G+15-storey RCC residential tower with a shear wall core designed for Chandigarh, India. Applying the comprehensive LCA methodologies taught in the Sustainable Engineering and Circular Economy course, the study covers the full building life cycle from raw material extraction through construction, a 50-year operational phase, and end-of-life demolition. Quantities are derived directly from the structural design calculations, ensuring a highly accurate, project-specific material inventory.

Key Findings:

- Total Global Warming Potential (GWP) over 50 years: ~43,990 t CO₂eq
- Embodied carbon (materials, A1–A3): ~3,613 t CO₂eq (~8.2% of lifecycle total)
- Operational carbon dominates at ~90% of total lifecycle emissions
- Embodied carbon intensity: ~484 kg CO₂eq/m² (in line with typical Indian high-rises)
- Fly ash bricks reduce masonry emissions significantly vs. clay bricks
- End-of-life steel recycling provides a credit of ~-301 t CO₂eq

1. Goal and Scope Definition

1.1 Goal

The goal of this LCA is to quantify the environmental burdens associated with the design, construction, use, and end-of-life of the residential tower at Chandigarh, using material quantities derived from the structural design calculations. This assessment strictly applies the principles of Life Cycle Assessment and circular economy frameworks as taught by **Prof. B.K. Dubey**. The study informs design-stage decisions on material selection, identifies environmental hotspots, and establishes a baseline for future design optimisation.

1.2 Functional Unit & System Boundary

The functional unit is: one residential tower providing 75 apartments (total built-up area 7,472.7 m²) over a service life of 50 years in Chandigarh, India. The study follows a cradle-to-grave system boundary aligned with EN 15978 / ISO 14044 modules, encompassing A1-A3 (Manufacturing), A4-A5 (Construction), B1-B7 (Use phase), C1-C4 (End-of-life), and D (Recycling benefits).

2. Life Cycle Inventory (LCI)

2.1 Building Data & Material Quantities

The following project parameters are extracted directly from the structural quantity takeoff:

Material / Component	Quantity	Unit	Embodied Carbon Factor	GWP (t CO ₂ eq)
M30 Concrete	4,937	m ³	330 kg CO ₂ eq/m ³	1,629.2
Fe500 Reinforcement Steel	543	MT	1,850 kg CO ₂ eq/MT	1,004.6
Fly Ash Brickwork	1,954	m ³	200 kg CO ₂ eq/m ³	390.8
Cement Plaster	16,281	m ²	20 kg CO ₂ eq/m ²	325.6
Ceramic Tile Flooring	6,512.7	m ²	35 kg CO ₂ eq/m ²	228.0

2.2 Justified Assumptions

Operational energy intensity is assumed at 130 kWh/m²/year based on the ECBC-R benchmark for the composite climate of Chandigarh (Zone 4). Grid carbon intensity is calculated using the CEA national average of 0.82 kg CO₂/kWh. Water consumption is derived directly from the plumbing network analysis (81,000 L/day).

3. Life Cycle Impact Assessment (LCIA)

3.1 Global Warming Potential (GWP)

Embodied Carbon (A1-A3): Concrete represents 45.1% of the material footprint, while reinforcement steel represents 27.8%. The total embodied carbon is 3,613.3 t CO₂eq, resulting in an intensity of 484 kg CO₂eq/m², which sits optimally within the standard Indian high-rise benchmark (350-600 kg CO₂eq/m²).

Operational Phase (B6): Over the 50-year design life, energy consumption generates 39,830 t CO₂eq, accounting for 90.5% of the total lifecycle emissions. Maintenance (B2-B5) contributes a marginal 112.7 t CO₂eq.

3.2 Life Cycle GWP Summary

Life Cycle Stage	Module	GWP (t CO ₂ eq)	% of Total
Raw material & manufacturing	A1–A3	3,613.3	8.2%
Transport & construction	A4–A5	144.7	0.3%
Maintenance & repair	B1–B5	112.7	0.3%
Operational energy (50 yr)	B6	39,830.0	90.5%
Demolition & disposal	C1–C4	348.5	0.8%
TOTAL	A + B + C	44,049.2	100%

4. Sensitivity Analysis & Hotspots

4.1 Grid Decarbonisation & Solar Integration

India's grid carbon intensity is projected to decline. A moderate decarbonisation scenario (0.50 kg CO₂/kWh by 2035) reduces B6 emissions by 35.3%. Furthermore, integrating rooftop solar PV on the expansive roof plan (particularly around the water tank zones) presents a highly feasible reduction strategy.

4.2 Material Selection Verification

The structural decision to utilise fly ash bricks instead of conventional clay bricks successfully avoided 351.7 t CO₂eq (a 90% reduction in masonry emissions), demonstrating active sustainable engineering principles applied during the design phase.

5. Conclusion & Recommendations

This cradle-to-grave LCA quantifies the environmental footprint of the B+G+15 residential tower. The total lifecycle GWP is approximately 43,990 t CO₂eq over 50 years. From a structural design perspective, the project already incorporates environmentally positive decisions, including the use of fly ash bricks and a highly optimised isolated footing design (4.2m x 4.2m) suited for the 100 kN/m² soil bearing capacity. To further optimise the design, it is recommended to specify blended cements (PPC/PSC with 40% fly ash) for the M30 concrete mixes and procure high-scrap EAF steel, which could reduce the embodied footprint by an additional 15-20% without requiring structural redesign.